

DANILO F. LLAGA JR.

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WORK EXPERIENCE

WISTRON INFOCOMM PHILIPPINES, Software/Automation Engineer Jun 2025 - Present.

Spearheaded key initiative projects by developing software solutions that helped improve efficiency in the production and human resources processes.

- **AI Chatbot Agent for Repair Engineers:** Implemented a RAG-based knowledge retrieval system using LangChain, ChromaDB vector database, and Ollama Mini 3 LLM to answer repair queries from 30,000+ historical records (50 columns). Accelerated onboarding for newly hired engineers by providing instant access to procedural knowledge and troubleshooting guidance, reducing dependency on senior engineer escalations for routine repair tasks.
- **Automated Optical Inspection (X-ray AOI):** Architected and deployed a PyTorch YOLO-based deep learning system to detect and classify defects on BGA components from live X-ray image feeds. Trained on 1,000+ annotated images across multiple defect types, achieving 98% detection accuracy. Implemented real-time operator validation workflow and web-based reporting dashboard, reducing manual inspection time by 40% and enabling quality engineers to process defects as the X-ray feed updates dynamically.
- **Facial Recognition DTR:** Built and deployed an InsightFace-based attendance system on Jetson Orin NX serving 500+ employees, logging time-in/time-out events to an HR web portal with 100% detection accuracy. Implemented as a backup layer on top of existing RFID card-swipe system, eliminating attendance gaps for employees who forget to swipe and automating HR reconciliation workflows that previously required manual intervention.
- **GPU SMX Socket Inspection:** Developed a YOLOv11-based computer vision pipeline to detect bent pins across 100-300 pin sockets, computing pixel-level bend percentages against quality acceptance criteria for automated defect classification. Achieved 96% accuracy, replacing time-consuming manual inspection that required operators to identify sub-millimeter defects across hundreds of pins per socket, significantly reducing operator fatigue and inspection bottlenecks.
- **Secure Test-Access System:** Developed a secure authentication and authorization system for whole-unit and motherboard repair workflows, ensuring controlled access to sensitive testing procedures and equipment.
- **Server Infrastructure and Distributed Computing:** Repurposed 3 unused PCs into production servers hosting multiple internal web applications (RAG AI Chatbot, GPU Pin Inspection webapp). Architected a distributed PyTorch training environment utilizing 4 GPUs across networked computers, enabling parallel model training and maximizing compute resource utilization for machine learning development.
- **Thermal-Imaging Device Usage Analytics:** Designed and implemented a logging system with HTML/CSS/JS web dashboard for a thermal-imaging short locator device, tracking usage patterns across 50+ daily users. Dashboard analytics identified the most frequently overheating components, providing data-driven insights that convinced management to expand thermal imaging capabilities to additional production lines.
- **Solder-Paste Applicator Quality Control:** Implemented a YOLOv11-based defect detection system for a semi-automatic solder-paste applicator, achieving 97% accuracy in identifying misalignment, over-application, and insufficient paste on individual solder balls. Integrated user activity logging and web-based defect tracking, automating the solder pasting quality assurance process and reducing operator touch time by 30%.

DOST-PES/SEI (Project LODI) WFH Setup Web Design Intern Jun - Aug 2024

Crafted interactive UI mockups and high-fidelity prototypes for a Project Portfolio Management website using Figma, aligning layouts with DOST branding and ensuring a mobile-first, responsive design.

- Built dynamic project dashboards in Google Data Studio to visualize key metrics (timelines, budget vs. actuals, and completion status), enabling stakeholders to track portfolio health at a glance.

PHILIPPINE COASTAL STORAGE AND PIPELINE CORP. Subic Bay Freeport Zone Accounting Office Intern Jul – Aug 2023

Assisted accountants in preparing for annual audits by meticulously arranging and documenting the company's payment vouchers.

- Provided invaluable support by fulfilling various tasks including printing, scanning, and organizing essential financial documents.
- Demonstrated strong attention to detail and organizational skills, contributing to the efficiency of the audit preparation process.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING at President Ramon Magsaysay State University, Iba, Zambales. July 2025. Software Development as Elective Subject

- Cumulative GPA: 1.53
- Relevant Coursework: Machine Learning, Microcontrollers/Microprocessors, Electronics, Embedded Systems, Web

THESIS

RICE LEAF DISEASE DETECTOR DEVICE WITH DENSENET-121 May 2025 *Thesis Project* | Machine Learning & Embedded System

- Designed and implemented a handheld rice-leaf disease detection device using Raspberry Pi 4B and a modified DenseNet-121 model, achieving up to 98% accuracy on the expert-validated Zambali dataset of 989 images and delivering real-time offline inference via a 7" touchscreen interface.
- Conducted comparative analysis of model performance across Zambali, Kaggle, and Hugging Face datasets, evaluating accuracy, precision, recall, and F1-score, demonstrating high improvement over baseline models and informing future dataset integration strategies.

ACTIVITIES

INSTITUTE OF COMPUTER ENGINEERS OF THE PHILIPPINES STUDENT EDITION PRMSU Iba President 2023-2024

- Authored and presented project proposals to secure funding and generate income for the organization, enabling the launch of the inaugural “PRMSU 1st ICpEP Challenge” and other signature events.
- Organized and led member meetings and coordinated all logistics for industry site visits, liaising directly with HR representatives at various companies to arrange educational tours and networking opportunities.

ADDITIONAL

Technical Skills: Arduino · Raspberry Pi · Jetson · Python · Java · Pytorch · Ultalytics YOLO · OpenCV · Node.js · HTML/CSS/JS · Figma · Git/GitHub. Also proficient in using Adobe Photoshop, Premiere Pro, & After Effects and various Microsoft Office Applications.

Languages: Fluent in Filipino; Conversational Proficiency in English; currently learning French (beginner level)

Certifications & Training:

Civil Service Certification - Professional

Safety Officer Level 2 Certification – Completed 240 hours of safety training

4th International Conference, 43rd Annual National Convention for Campus Student Advisers, and 27th Annual National Convention for Student Leaders - Baguio City

DOST-SEI Scholars Leadership Camp (2024) – Hilton Clark Sun Valley Resort, Pampanga

Awards:

Best Thesis Project

Best in Programming

Awarded Most Active Student Organization (ICpEP.se-PRMSU Chapter)

2nd Runner Up: 8th Regional CpE Challenge (Quizbowl)

3rd Runner Up: 7th Regional CpE Challenge (Python Programming)

Character references can be provided upon request.